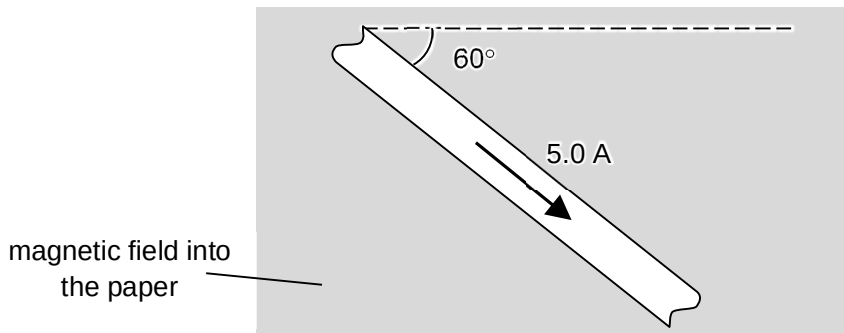


$$I_{\text{total}} = 1200 + 1700 = 2900 \text{ A}$$

- 22** A wire of length 3.0 cm is placed in the plane of the paper, along a line 60° clockwise from the x -axis. A magnetic field of flux density 0.040 T acts into the paper. The wire carries a current of 5.0 A.



What is the magnitude of the force which the field exerts on the wire?

- A** 0.0060 N **B** 0.0030 N **C** 0.0052 N **D** 0.0104 N

Ans: (A)

$$F = BIL = (0.040)(5.0)(0.030) = 0.0060 \text{ N}$$

The angle does not play a part.